CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 16: Loci (Locus and its Constructions)

Topic-wise short notes with solved examples

A locus is the path traced by a point that moves so as to satisfy a given condition. This chapter covers the standard loci, the two locus theorems and the constructions based on them.

1 What a locus is

- The **locus** of a point is the set of all points, and only those points, that satisfy a given condition. The plural is **loci**.
- To prove that a line or curve is a locus, two things must be shown: (i) every point on it satisfies the condition, and (ii) every point that satisfies the condition lies on it.
- A locus is usually a line, a pair of lines or a circle.

2 The two locus theorems

- **Theorem 1**: the locus of a point equidistant from two given points is the **perpendicular bisector** of the line segment joining them.
- **Theorem 2**: the locus of a point equidistant from two intersecting lines is the pair of **bisectors of the angles** between the lines.
- Converses: every point on the perpendicular bisector of AB is equidistant from A and B; every point on the bisector of an angle is equidistant from the arms of the angle.

Example 2.1 P is a point on the perpendicular bisector of AB. If $PA = 5$ cm, find PB. Give a reason.

SOLUTION

Every point on the perpendicular bisector of a segment is equidistant from its end points.

$PB = 5$ **cm**

3 Standard loci

Condition on the moving point	Locus
Equidistant from two fixed points A and B	Perpendicular bisector of AB
Equidistant from two intersecting lines	The pair of bisectors of the angles between them
Equidistant from the two arms of an angle	The bisector of the angle
At a fixed distance r from a fixed point O	Circle with centre O and radius r
At a fixed distance d from a given line	A pair of lines parallel to it, one on each side, at distance d
Equidistant from two parallel lines	The line parallel to both and midway between them
Equidistant from two concentric circles	The concentric circle midway between them
Mid-points of equal chords of a circle	A concentric circle
Centres of circles touching a given line at a given point	The line perpendicular to the given line at that point

Example 3.1 Describe the locus of (i) the centre of a wheel of radius 30 cm rolling along a straight road (ii) the tip of the minute hand of a clock.

SOLUTION

(i) The centre is always 30 cm above the road.

(ii) The tip is always at a fixed distance (the length of the hand) from the centre of the clock.

(i) A line parallel to the road, 30 cm above it. (ii) A circle whose centre is the centre of the clock and whose radius is the length of the minute hand.

Example 3.2 Describe the locus of points that are 3 cm from a fixed line AB.

SOLUTION

A point at a fixed distance from a line can lie on either side of it.

A pair of lines parallel to AB, one on each side, each at a distance of 3 cm from it.

4 Loci in a triangle

- The point equidistant from the three **vertices** of a triangle is the **circumcentre**. It is where the perpendicular bisectors of the sides meet.
- The point equidistant from the three **sides** of a triangle is the **incentre**. It is where the bisectors of the angles meet.

Example 4.1 In $\triangle ABC$, a point P is equidistant from A and B and also equidistant from AB and BC. How is P located?

SOLUTION

Equidistant from A and B: P lies on the perpendicular bisector of AB.

Equidistant from AB and BC: P lies on the bisector of $\angle ABC$.

P is the point where the perpendicular bisector of AB meets the bisector of $\angle ABC$.

5 Construction questions

- Construct each locus separately, using ruler and compasses only, and leave the construction arcs visible.
- A point that satisfies two conditions is the **intersection** of the two loci.
- Measure and record whatever the question asks for at the end.

Example 5.1 Construct $\triangle ABC$ with $AB = 6$ cm, $BC = 5$ cm and $\angle ABC = 60^\circ$. Locate the point P inside the triangle that is equidistant from B and C and also equidistant from AB and BC. Write the steps.

SOLUTION

Step 1: draw $BC = 5$ cm. At B construct an angle of 60° and cut off $BA = 6$ cm. Join AC.

Step 2: draw the perpendicular bisector of BC. This is the locus of points equidistant from B and C.

Step 3: draw the bisector of $\angle ABC$. This is the locus of points equidistant from AB and BC.

Step 4: the point where the two loci meet is the required point P. Measure PB if asked.

Example 5.2 Points A and B are 6 cm apart. Describe how to find the points that are 4 cm from A and equidistant from A and B.

SOLUTION

Locus 1: the circle with centre A and radius 4 cm.

Locus 2: the perpendicular bisector of AB.

The circle and the bisector meet at two points; these are the required points.

6 Common mistakes

- Giving one parallel line instead of a **pair** for points at a fixed distance from a line.
- Giving one angle bisector instead of the **pair** for two intersecting lines.
- Confusing the circumcentre (perpendicular bisectors of sides) with the incentre (angle bisectors).
- Rubbing out the construction arcs.